

The band at a local school cannot attend all of the games for all sports teams in the month of March due to conflicting schedules. The Athletic Director, Ms. Cotton, wants to prioritize the games the band will perform to boost morale of the teams. Ms. Cotton asks 250 students to pick the sport the band team should play. The results were recorded in the table below. The school has a total of 2500 students. Use the results of the sample to answer questions 1-5.

1. Complete the table with ratios of sports picked. Show your work.

Sport	Number of Times Chosen	Ratio of Students That Picked Each Sport
Baseball	32	$\frac{32}{250} = \frac{16}{125}$
Basketball	125	$\frac{125}{250} = \frac{1}{2}$
Track	43	$\frac{43}{250}$
Volleyball	50	$\frac{50}{250} = \frac{1}{5}$

2. Predict the number of students in the school that want the band to perform at the basketball games (based on sample results) using the percent method.

1) $\frac{1}{2} = 0.5 = 50\%$ 2) $0.5 (2,500) = 1,250$ students

3. Explain which is the best method to predict the number of students in the school that want the band to perform at the track games (based on sample results)?

Equivalent fractions because $\frac{2500}{250} = 10$ $\frac{43 \cdot 10}{250 \cdot 10} = \frac{430}{2500}$ 430 students

4. Ms. Cotton uses the scale factor method to predict the number of students in the school that want the band to play at volleyball games. She uses a scale factor of 0.2. Did Ms. Cotton use the right scale factor? Explain.

She did not. To find scale factor use $\frac{2500}{250} = 10$

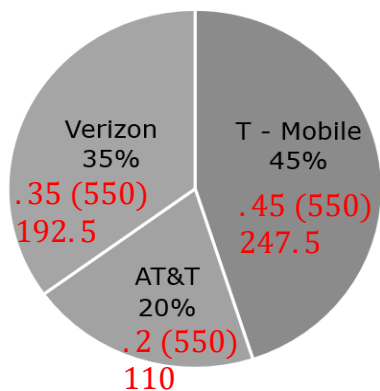
5. The band prefers to play at the outdoor sports. Predict the amount of students that prefer the band to perform at the outdoor sports.

Baseball + track
 $\frac{32+43}{250} = \frac{75}{250} = 0.3$ $0.3 (2,500) = 750$ students

550 students in the 9th grade were polled to see what cell phone carrier was the most common, in a local high school of 2,475 students. The circle graph below shows the percentage of students that have the three major cell phone carriers: AT&T, T- Mobile and Verizon.

Use the circle graph to answer questions 6-8.

Student Cell Phone Carriers



6. To predict how many students in the school use AT&T as a carrier, the student council president wants to use the scale factor method. What scale factor should she use to make this prediction?

$$\frac{2,475}{550} = 4.5$$

7. Explain whether the equivalent fraction or percent method be used to predict the number of students with Verizon as a carrier.

Percent method since percents are given in circle graph.

8. Predict the number of students that have T-Mobile as a cell phone carrier.

$$.45 (2475) \approx 1,113.75 = \text{About } 1,114 \text{ students}$$

Leroy wants to know about the color distribution of gummy worms in his candy jar. Without looking he picked out one gummy worm at a time, recorded it, and put them back into the jar. Leroy picked 12 green, 16 red, 24 yellow, 8 orange after 60 picks. A total of 260 gummy worms are in the jar. Use Leroy's results to answer questions 9-10.

9. What fraction of gummy worms are ¹²green, ¹⁶red, and ⁸orange?

$$\frac{36}{60} = \frac{18}{30} = \frac{9}{15}$$

10. Assuming his sample is a true representation of gummy worm distribution, how can Leroy predict the estimated number of gummy worms in the jar that are yellow?

1) Find percent that are yellow in his sample: $\frac{24}{60} = 0.4 = 40\%$

2) Multiply the 40% by total of 260 gummy worms.

$$0.4 (260) = 104 \text{ gummy worms}$$